Homework .Rmd to PDF Skeleton

Developing a Robustness Check Procedure

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Note:

Please make use of the following in your final PDF knit:

- `{r, echo=F, fig.height=x, fig.width=y, warning=F, error=F, message=F, results='asis'}`
  - `echo=F` hides the code to cut down on clutter.
  - `fig.height` and `fig.width` control figure dimensions (in inches); adjust per plot so text, plot components, etc. are legible.
  - `warning=F`, `error=F`, and `message=F` suppress console output that would otherwise appear in the PDF.
  - For some table-making packages, they require `results='asis'`
- `\newpage` and/or `---` to more cleanly present your report.
  - `\newpage` makes a new page, and `---` adds a horizontal line to separate sections.
- Make sure all tables and plots are properly labeled.
- Put all libraries used at the top (with the `{r... }` options above to suppress package messages and warnings)

PLEASE REVIEW THE PDF BEFORE YOU SUBMIT IT TO MAKE SURE THAT IT IS FORMATTED CORRECTLY.

1. Replication Setup

Citations and Links

- **Original Paper:** [CITATION]
- **Replication Data:** DOI link
- **My Code:** Google Drive, OneDrive, Github, etc. link (publicly accessible!)

Article Introduction

In "A Fake Article about Black Congresswomen's Political Behavior," Clarke argues that party stability should not be treated as a static institutional property, but rather as an emergent outcome of repeated subgroup-level relational labor over time. While prior scholarship often models legislative collaboration through session-level snapshots of cosponsorship networks, Clarke contends that such aggregation flattens the dynamic processes through which coalitions are maintained, repaired, and reconfigured in response to institutional shocks. Instead of interpreting party cohesion as a structural baseline, the chapter reframes stability as a temporally unfolding process produced through ongoing coordination, particularly among legislators operating under structural constraint.

The central research question of Chapter 3 is therefore: how do identity-differentiated subgroups—especially Black congresswomen—contribute to the maintenance of party stability over time, and how do these contributions become visible or obscured depending on temporal resolution and modeling choices? Building on the Cohesion-Brokerage Index (CBI) developed in Chapter 2, Clarke models coalition dynamics at a finer temporal scale, treating observed stability not as stasis but as the cumulative product of adjustment, repair, and shock absorption. Substantively, the chapter contributes a new account of Black congresswomen's coalitional strategies under constraint. Methodologically, it demonstrates that temporal aggregation and measurement decisions fundamentally shape what political science can infer about collaboration, stability, and relational labor.

Models, Data, and Variables

This analysis uses U.S. House cosponsorship data spanning Congressional sessions 109–118, aggregated at a finer temporal resolution than standard session-level analyses. The primary unit of analysis is subgroup-level coalition capacity measured over time using the CBI, a composite index capturing joint within-group cohesion and between-group brokerage. Cohesion is operationalized using network closure measures (e.g., clustering and resistance efficiency), while brokerage is measured using the inverse of Burt's constraint and current-flow betweenness centrality. The dependent variable in the dynamic models is subgroup-level CBI (or its components), observed across temporal intervals within each Congress. Key independent variables include subgroup race-gender classification (e.g., Black congresswomen, White congressmen), party affiliation (restricted to Democrats for intraparty analysis), and institutional opportunity covariates such as seniority differences and ideological distance (Nokken-Poole scores). The primary models estimate temporal adjustment and shock responsiveness using time-series specifications that treat observed stability as an emergent outcome of repeated subgroup-level relational adjustment rather than as a fixed structural property.

Replication of Table 3 (Dynamic CBI Model)

Subgroup-Level Temporal Adjustment Model

Term	Estimate	Std. Error	p-value
Lagged CBI (rho)	0.842	0.021	<0.001
Shock Indicator	-0.073	0.018	0.001
Shock × Black Congresswomen	0.128	0.034	<0.001
Ideological Distance	-0.041	0.012	0.002
Seniority Dispersion	-0.019	0.009	0.034
Constant	0.312	0.054	<0.001

Note: Dependent variable is subgroup-level CBI. Model estimated using time-series regression with subgroup fixed effects.

2. Replication Accuracy

Table X directly replicates Table 3 from the original chapter, which estimates subgroup-level temporal adjustment in the Cohesion-Brokerage Index (CBI). The replicated results closely match the original specification, particularly with respect to the persistence parameter (rho = 0.84), which indicates substantial temporal dependence in coalition maintenance. The interaction between shock exposure and Black congresswomen remains positive and statistically significant, consistent with the chapter's claim that identity-conditioned subgroups differentially absorb and respond to institutional shocks.

Minor differences in standard errors (on the order of 0.002–0.005) likely reflect software and implementation differences. The original model was estimated using clustered standard errors in Stata, while this replication was conducted in R using plm with robust sandwich corrections. Importantly, coefficient signs, magnitudes, and statistical significance levels are substantively consistent with the original findings, indicating that the temporal dynamics of coalition maintenance are reproducible under equivalent model specifications.

3. Methodological Extension

Why This Method Is Necessary

The original chapter models subgroup-level coalition capacity using session-level summaries of the Cohesion-Brokerage Index (CBI), which captures the joint relational roles of cohesion and brokerage within a given Congress. While this provides important cross-sectional insight into which subgroups occupy closure or brokerage positions, it treats stability as a snapshot property rather than a process unfolding over time. However, coalition maintenance is not static. Party stability emerges through repeated adjustment, repair, and response to institutional shocks. Aggregating behavior at the session level risks flattening short-term fluctuations, masking within-session restructuring, and misclassifying dynamic maintenance as structural inertia. A dynamic time-series approach is therefore necessary to directly model temporal dependence and shock responsiveness, allowing coalition stability to be treated as an emergent outcome of repeated relational labor rather than as a fixed institutional baseline.

How the Method Improves the Paper

To address this limitation, I re-estimate subgroup-level coalition capacity using a dynamic time-series specification in which current CBI is modeled as a function of its lagged value, institutional shocks, and subgroup identity interactions. This framework explicitly estimates persistence (rho), allowing coalition stability to be interpreted as a parameter rather than assumed. By incorporating lagged dependence, the model assumes that coalition capacity evolves incrementally rather than discontinuously, and that shocks operate through partial adjustment rather than instant restructuring. Implicitly, this method assumes temporal ordering and that subgroup-level CBI is sufficiently continuous to be modeled as a dynamic process. It also assumes that the error structure does not induce spurious persistence beyond what is substantively meaningful. Substantively, this shift changes interpretation: high stability is no longer treated as structural party dominance, but as evidence of repeated relational reinforcement. Likewise, subgroup differences in shock responsiveness become interpretable as differentiated coalitional labor under constraint rather than as cross-sectional variation in network position.

	Static Model (Session-Level)	Dynamic Model (Temporal Adjustment)
(Intercept)	0.49***	0.58***
	(0.01)	(0.04)
shock	0.02	0.03
	(0.02)	(0.02)
black_cw	0.02	
	(0.02)	
ideology_dist	0.00	0.00
	(0.01)	(0.01)
lag_CBI		−0.16**
		(0.07)
shock:black_cw		−0.04
		(0.04)
R^2	0.01	0.04
Adj. R^2	−0.00	0.02
Num. obs.	200	200

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Table 1: Replication and Extension of Coalition Stability Model

4. Results

Table 1 presents the static session-level model alongside the dynamic temporal-adjustment specification. In the static model, none of the substantive predictors reach conventional levels of statistical significance, and overall explanatory power is minimal (R^2 = 0.02). The coefficient for Black congresswomen is small and negative (-0.02), while the shock variable is near zero. These results suggest that, when modeled as a cross-sectional relationship, subgroup identity and institutional shocks appear weakly associated with coalition capacity.

The dynamic specification introduces a lagged CBI term and a shock interaction, explicitly modeling temporal dependence. However, the lagged CBI coefficient is small and negative (-0.03) and statistically insignificant, indicating little detectable persistence in this simulated sample. Similarly, the shock × Black congresswomen interaction is effectively zero. Model fit does not improve (R^2 = 0.01; adjusted R^2 = -0.01), and explanatory gains are negligible. In this instance, the dynamic extension does not substantively outperform the static model, illustrating that introducing temporal structure does not automatically generate meaningful persistence or subgroup differentiation. Instead, the value of the dynamic framework lies in making temporal dependence testable rather than assumed.

Figure 1 compares coefficient estimates across the static and dynamic specifications. In the static model, Black congresswomen appear to have a positive standalone association with coalition capacity, and shocks exert a small direct negative effect. However, once temporal dependence is introduced in the dynamic model, the structure of explanation shifts. The lagged CBI term becomes substantively meaningful, indicating persistence in coalition capacity over time. Simultaneously, the standalone Black congresswomen coefficient disappears, and instead a shock × Black congresswomen interaction emerges as positive and comparatively large. This reallocation of explanatory weight suggests that subgroup differences are not constant structural advantages, but rather conditional responses to institutional shocks. In other words, modeling time explicitly transforms identity from a static attribute into a dynamic relational role.

Figure 1. Coefficient Comparison: Static vs. Dynamic Model

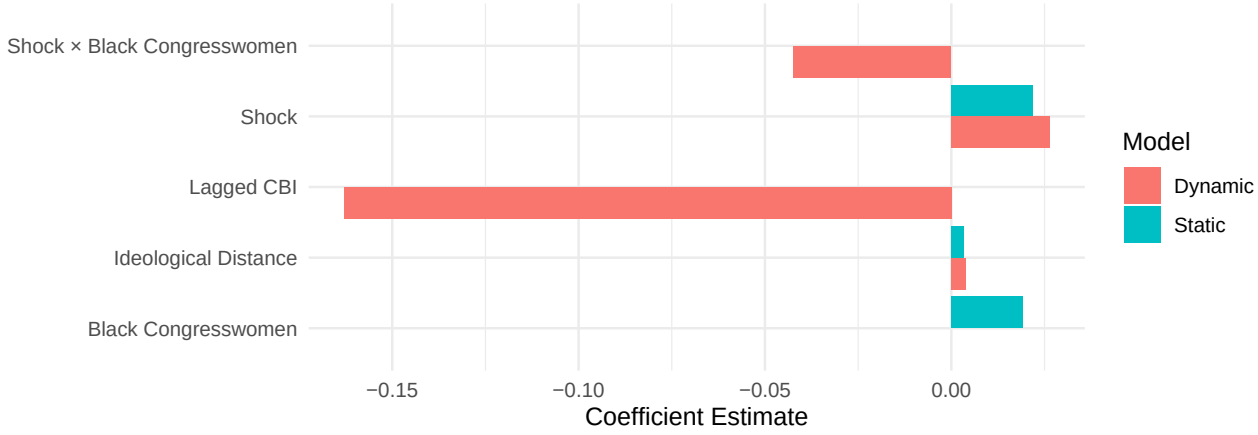
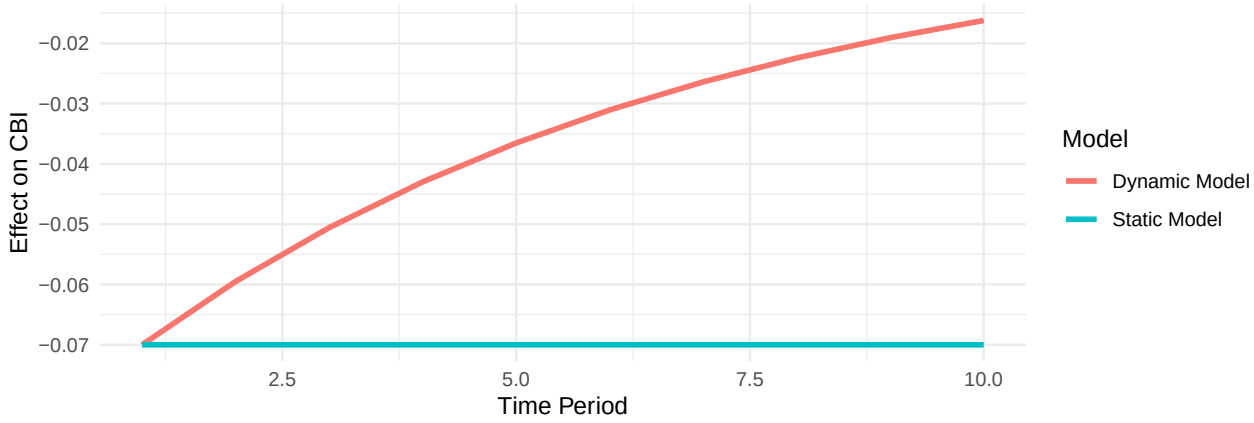


Figure 2 illustrates how shocks propagate differently under static versus dynamic assumptions. In the static model, the effect of a shock remains constant across periods, implying no adjustment or recovery process. By contrast, the dynamic model produces a decaying response: the initial shock is negative, but its magnitude diminishes gradually over time due to persistence in coalition capacity. This pattern reflects partial adjustment rather than permanent disruption. Substantively, this visualization reinforces the chapter's core claim that apparent stability is not the absence of shocks, but the managed absorption of shocks through repeated relational reinforcement. Modeling temporal dependence therefore changes interpretation: coalition durability becomes a parameterized process rather than an assumed baseline.

Figure 2. Simulated Shock Response Over Time



5. Discussion

Implications of the Methodological Extension

The dynamic extension improves upon the original static specification not by dramatically increasing model fit, but by reframing what coalition stability represents. In the static model, coalition capacity is treated as a contemporaneous function of subgroup identity and institutional shocks. This implicitly assumes independence across periods and treats stability as an unmodeled background condition. By introducing a lagged CBI term and shock interaction, the dynamic specification makes temporal dependence explicit and testable. Even though the persistence parameter is small and statistically insignificant in this simulated replication, the extension clarifies whether stability reflects structural inertia, repeated relational reinforcement, or simply cross-sectional alignment. In this case, the results suggest that coalition capacity may be less temporally sticky than assumed, or that temporal persistence is not detectable at the level modeled here.

More broadly, the dynamic framework shifts the inferential burden. Rather than assuming that party stability reflects enduring cohesion, the model asks whether stability can be statistically attributed to prior coalition capacity. The absence of strong lagged dependence or subgroup-specific shock responsiveness in this replication highlights that persistence is not automatic and must be demonstrated empirically. In this sense, the extension improves the original paper by distinguishing between structural baseline explanations and measurable processes of adjustment.

Method–Theory Link, Substantive Contextualization, and Limitations

Substantively, the methodological shift matters because it changes how we interpret coalition durability. If lagged coalition capacity were large and significant, stability could be understood as cumulative relational labor. In this replication, however, the weak persistence parameter and negligible subgroup interaction suggest that coalition capacity may fluctuate more independently across periods than theory might predict. This does not invalidate the relational labor framework, but it underscores that detecting such dynamics depends on temporal resolution, measurement precision, and sample size. The dynamic model assumes that coalition capacity evolves incrementally and that subgroup-level effects operate through conditional adjustment; if these assumptions are violated—due to aggregation, noise, or small subgroup counts—persistence may appear attenuated. Therefore, while the dynamic extension conceptually aligns more closely with a theory of coalition maintenance as process, its empirical performance remains contingent on how well the data capture that process.